LUC BROGAT-MOTTE

Postdoctoral researcher in machine learning ◇ luc.brogat.motte@gmail.com

EXPERIENCE

Postdoctoral Researcher, Istituto Italiano di Tecnologia *2025–*
Supervised by Lorenzo Rosasco (MaLGa, University of Genoa).
Research on the estimation of continuous-time stochastic dynamical systems.

Postdoctoral Researcher, CentraleSupélec *2023–2025*
Supervised by Riccardo Bonalli (L2S, CentraleSupélec) and Alessandro Rudi (SIERRA, INRIA Paris).
Research on the estimation of controlled stochastic differential equations.

Research Intern (Master's), Télécom Paris & Aalto University *Apr.–Sept. 2019*
Learning output representations in structured prediction with applications to molecular prediction.

EDUCATION

PhD in Mathematics and Computer Science, Télécom Paris *2019–2023*
Supervised by Florence d'Alché-Buc (Télécom Paris) and Juho Rousu (Aalto University).
Research topics: statistical learning theory, kernel methods, optimal transport.
Runner-up for the Best PhD Thesis Award (CSDAI, Institut Polytechnique de Paris), 2024.

ENS Paris-Saclay *2018–2019*
M.S. MVA (Mathématiques, Vision, Apprentissage), graduated with honors (16.8/20).

Télécom Paris *2016–2019*
Engineering degree in applied mathematics (probability, statistics, scientific computing), GPA: 3.98/4.

CPGE MPSI/MP* — Lycée Marcelin Berthelot *2013–2016*
Intensive undergraduate program in mathematics and physics.

PUBLICATIONS AND PREPRINTS

Markov chain convergence in kernel-induced geometry. (Preprint)
Oleksii Kachaiev, Luc Brogat-Motte, Silvia Villa, Lorenzo Rosasco.

Fast learning rates for physics-informed kernel methods. (NeurIPS 2026)
Luc Brogat-Motte, Joachim Bona-Pellissier, Giacomo Meanti, Lorenzo Rosasco.

Learning controlled stochastic differential equations. (Preprint)
Luc Brogat-Motte, Riccardo Bonalli, Alessandro Rudi.

Safely learning controlled stochastic dynamics. (NeurIPS 2025)
Luc Brogat-Motte, Alessandro Rudi, Riccardo Bonalli.

Sketch in, sketch out: accelerating both learning and inference for structured prediction with kernels. (AISTATS 2024)
Tamim El Ahmad, Luc Brogat-Motte, Pierre Laforgue, Florence d'Alché-Buc.

Learning to predict graphs with fused Gromov–Wasserstein barycenters. (ICML 2022)
Luc Brogat-Motte, Rémi Flamary, Céline Brouard, Juho Rousu, Florence d'Alché-Buc.

Vector-valued least-squares regression under output regularity assumptions. (JMLR 2022)
Luc Brogat-Motte, Alessandro Rudi, Céline Brouard, Juho Rousu, Florence d'Alché-Buc.

Duality in RKHSs with infinite-dimensional outputs: application to robust losses. (ICML 2020)
Pierre Laforgue, Alex Lambert, Luc Brogat-Motte, Florence d'Alché-Buc.

STUDENT SUPERVISION

PhD student (previously Master's research intern)
Luca Piccaluga, *Forecasting algorithms for continuous-time stochastic dynamical systems*.
Co-supervised with Lorenzo Rosasco. *June 2025–*

Master's research intern
Hubert Leroux, *A-PIKS: Active Learning for Physics-Informed Kernel Methods*.
Co-supervised with Joachim Bona-Pellissier and Lorenzo Rosasco. *Apr.–Aug. 2026*

Master's research intern
Florentin Acker, *Physics-Informed Kernel Learning with Misspecified Physical Constraints*.
Co-supervised with Joachim Bona-Pellissier and Lorenzo Rosasco. *Apr.–Aug. 2026*

TEACHING

Teaching assistant, University of Genoa *2025–2026*
20 hours (M2) in "Sequential Prediction and Reinforcement Learning". Contributed to lectures and practical sessions.

Teaching assistant, Télécom Paris *2019–2023*
130.5 hETD (L3–M2) in statistics and machine learning. Courses include: Statistics (L3), Machine Learning (M1), Kernel Machines (M2), Structured Data: Learning and Prediction (M2), Structured Prediction (Executive MSc).

PRESENTATIONS

NeurIPS 2026, Paris, France *Dec. 2026*
Poster presentation.

WCCM-ECCOMAS 2026, Munich, Germany *July 2026*
Oral presentation, "Learning Controlled Stochastic Differential Equations" (MS237: Recent Advances in Scientific Machine Learning for Differential Equations and Control Problems).

Laboratoire de l'Informatique du Parallélisme (LIP), ENS de Lyon *Apr. 2026*
Invited research seminar.

Rencontres de Statistiques du CEREMADE, Université Paris-Dauphine *Dec. 2025*
Invited talk.

NeurIPS 2025, San Diego, USA *Dec. 2025*
Poster presentation.

Invited research talks *Sept.–Oct. 2025*
I2M Marseille (Statistics Seminar); IMT Toulouse (Statistics and Optimization Seminar);
LAAS-CNRS (POP team, Toulouse); GIPSA-Lab (GAIA team, Grenoble);
LIS (QARMA team, Marseille); Inria Rennes (MALT team).

Isaac Newton Institute, Cambridge, UK *July 2025*
Invited seminar, "Representing, calibrating & leveraging prediction uncertainty" workshop.

CSDAI Best PhD Thesis Awards *June 2024*
Presentation as recipient of the CSDAI Best PhD Thesis Award (runner-up).

ICML 2022, Baltimore, USA *July 2022*
5-minute oral and poster presentation.

ELLIS Theory Workshop, Arenzano, Italy *July 2022*
Poster presentation.

IDIA-IP Paris Day, École Polytechnique *June 2022*
Poster presentation.

UMR MIA Seminar, AgroParisTech *June 2022*
Research seminar talk.

LIKE22 Workshop *Jan. 2022*
Contributed talk.

ACADEMIC SERVICE

Member, COST Action CA24136 (InterCoML) *2025–*
European working group on interactions between control theory and machine learning.

Reviewer for international journals and conferences *2019–*
JMLR, Constructive Approximation, SIAM Journal on Mathematics of Data Science,
Artificial Intelligence Journal, Systems & Control Letters,
NeurIPS, AISTATS, ECML PKDD.
Top Reviewer, NeurIPS 2026.

Co-organizer of the MaLGa Seminar Series (15 seminars). *2025–*